

Next Generation Data Centers and Energy Management Systems

Lecture 5B: Liquid Cooling & Advanced Heat Transfer

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Learning Plan

- Introduction to Data Center Systems and Thermodynamics
- Psychrometrics and Heat Transfer
- Performance Metrics and Standards
- Applied Computational Fluid Dynamics (CFD)
- **Liquid Cooling & Advanced Heat Transfer**
- Chip Cooling: Heat Conduction and Thermal Interface Materials
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- Overview of Simulation Field; the Concept of a Digital Twin
- Machine Learning and Optimization Tools
- Modeling and Simulation of an Energy Management System
- Control Systems and Optimization
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- Explain the physical limitations of air cooling in high-thermal design power environments.
- Identify the key components used to transfer heat away from a chip (TIMs, spreaders, heat pipes).
- Differentiate between various liquid cooling architectures (Cold Plates vs. Immersion).
- Describe the role and function of a cooling distribution unit (CDU) in a data center environment.

Seminar 2

- Dr. David Geb, Synopsys (formerly Ansys)
- Title: TBD
- Date: Tuesday, 9/29/26
- Time: 15:30 (Berkeley Time)
- Location: GSPP 150

Typical layout of a data center facility

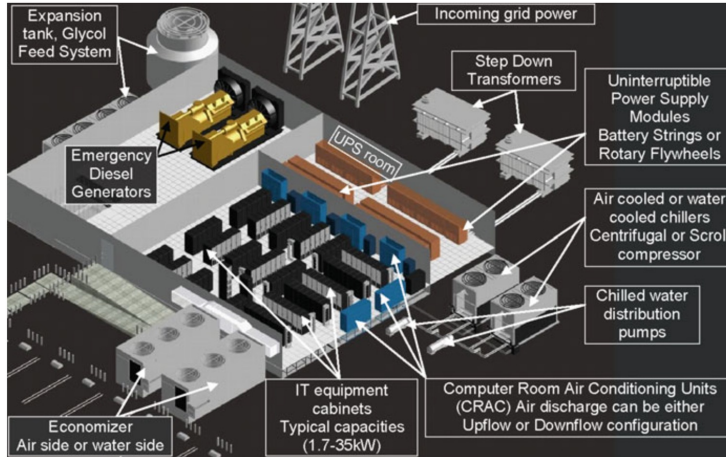
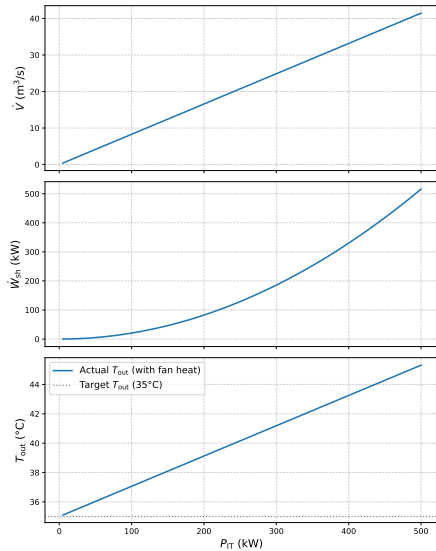


Figure 1: (Figure 1.1 in Ref. [2]).

Liquid cooling

- **Power Density:** Increasing TDP (thermal design power) leads to extreme heat fluxes at the chip level.
- **Limitations of Air:**
 - Low volumetric heat capacity.
 - Requires large volumes of air and high fan speeds (power intensive).
 - Limited heat transfer coefficient.
- **The Liquid Advantage:**
 - Much higher thermal conductivity and heat capacity.
 - Allows for much higher heat flux removal in a smaller footprint.

Fan Laws



Cooling technologies

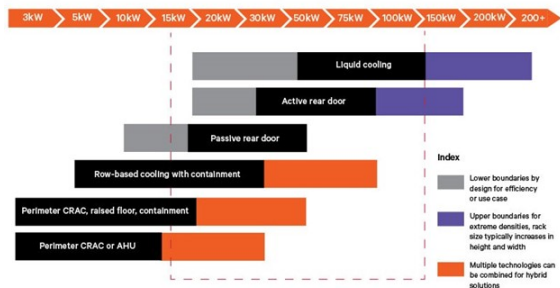


Figure 2: Understanding direct-to-chip cooling in HPC infrastructure: A deep dive into liquid cooling. Ref: [Vertiv](#).

Thermal Interface Materials (TIM)

- **The Problem:** Surface roughness at the chip/spreader interface creates microscopic air gaps (insulators).
- **The Solution:** TIMs (greases, pads, liquid metal) fill these gaps to create a continuous conduction path.
- **Key Metrics:**
 - **Thermal Conductivity (k):** Higher is better.
 - **Bond line thickness:** Minimizing thickness reduces total thermal resistance (R_{th}).

Heat Spreaders

- High-conductivity metal (usually copper) to distribute heat laterally.
- Reduces "spreading resistance."

Heat pipes and vapor chambers

- Use phase change (evaporation/condensation) to transport heat.
- Heat pipe: 1D transport via capillary action.
- Vapor chamber: 2D spreading via phase change; extremely efficient for high-flux chips.

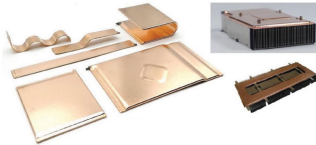


Figure 3: Various vapor chambers [1].

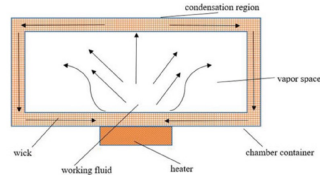


Figure 4: Structure of a vapor chamber (Schematic: dimensions and sizes are not to scale) [1].

Heat pipes and vapor chambers

	Vapor Chamber	Heat Pipe
Theory	Two-phase heat transfer	Two-phase heat transfer
Application	Two-dimensional heat distribution. Spreading heat by a single vapor chamber. Suitable for large heat flux and high power	One-dimensional heat distribution. Using one or more heat pipes to spread heat. Suitable for long distance between heat source and heat exchanger.
Shape	Complex shape in x and y direction with pedestal, flattened rectangle, surface embossing and z-direction bendable	Round, flattened or bent in any direction
Sealing method	Swaged and welded on each end	Pinched and welded on each end, four sided diffusion bonding or welding
Fixtures	Mounted with through-holes in vapor chamber	Additional fixture plates needed to mount heat pipes
Mounting to heat source contact	Direct contact	Indirect contact through base plate unless flattened and machined

Figure 5: A comparison between vapor chambers and heat pipes [1].

Liquid Cooling Architectures: Cold Plates

- **Direct-to-Chip (cold plate) Cooling:**

- A liquid-cooled metal block is mounted directly to the chip.
- Liquid flows through internal channels to absorb heat.
- [Video](#) of the “skiving” process.

- **Microchannel Technology:**

- Etching tiny channels into the cold plate to vastly increase the surface area.
- Maximizes the convective heat transfer coefficient (h).

Liquid Cooling Architectures: Immersion

- **The Concept:** Submerging the entire IT hardware in a dielectric (non-conductive) fluid.
- **Single-Phase Immersion:**
 - Fluid remains liquid; cooled via convection and circulated by pumps.
- **Two-Phase Immersion:**
 - Fluid boils at the chip surface.
 - Utilizes **latent heat of vaporization** for superior cooling.

System-Level: Cooling Distribution Unit (CDU)

- **The Role:** Manages the heat exchange between the facility water and the IT cooling loop.
- **Two-Loop System:**
 - **Primary Loop:** Facility water (chilled water from the building).
 - **Secondary Loop (TCS):** Treated/dielectric fluid that circulates through the servers.
- **Core Functions:**
 - Heat Exchange (Plate Heat Exchangers).
 - Pumping and Flow Control.
 - Filtration and Water Chemistry Management.
 - Leak Detection and Safety Monitoring.

System-Level: Cooling Distribution Unit (CDU)

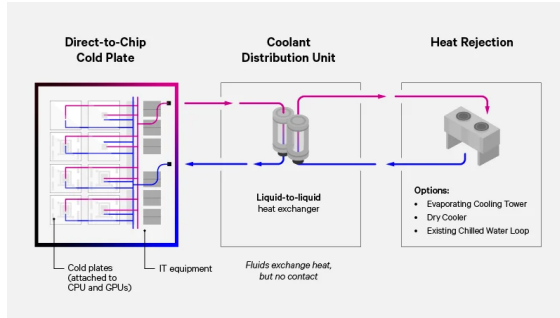


Figure 6: Vertiv.

Case Study: Aquasar Technology

- **Concept:** A hot water cooled supercomputer prototype designed for direct energy reuse [3].
- **Performance Comparison:**
 - **Liquid (Hot Water):** $\Delta T_{chip/coolant} \approx 15^{\circ}\text{C}$ (at 60°C coolant).
 - **Air Cooling:** $\Delta T_{chip/air} \approx 35^{\circ}\text{C}$ (requiring pre-cooling to 23°C).
- **Key Advantages:**
 - Eliminates the need for energy-intensive chillers.
 - Enables **Heat Reuse**: 60°C water can be used for building heating.
 - **Efficiency**: Achieved 80% heat recovery efficiency and 34% exergetic efficiency.
- **Economic Insight:** Space heating provides the highest economic value for recovered heat.

Case Study: Aquasar Technology

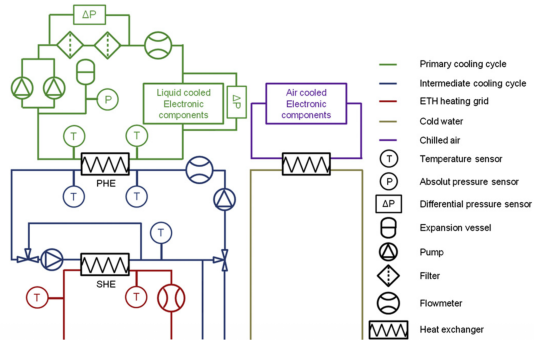


Figure 7: Aquasar: Schematics of the cooling loop [3]. Let us analyze the second law efficiency of the green loop.

Case Study: Aquasar Technology

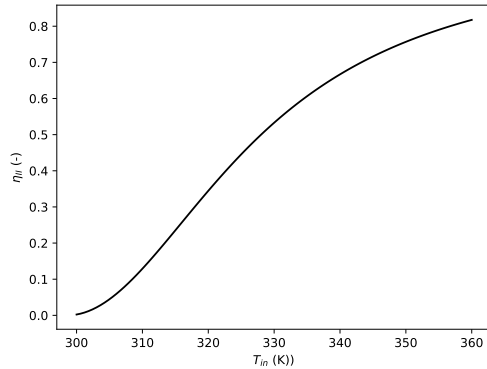


Figure 8: Second-law efficiency of the secondary TCS loop [3].

Radiative Cooling in Space Systems

- **The Challenge:** In the vacuum of space, there is no atmosphere for convection or conduction to a medium. **Radiation** is the only mechanism for heat rejection.
- **The Thermal Path:**
 - High-power chips → Liquid/Pumped Loops → External Radiator Panels → Deep Space.
- **Physics (Stefan-Boltzmann Law):**

$$Q_{rad} = \epsilon \sigma A (T_{rad}^4 - T_{space}^4)$$

- **Key Design Considerations:**
 - **High Emissivity (ϵ):** Using specialized coatings to maximize heat rejection.
 - **Radiator Area (A):** Large surface areas are required due to the T^4 dependence.
 - **Solar/Albedo Loading:** Managing heat from the Sun or Earth that hits the radiator.
 - **Mass Constraints:** Every kg of radiator/fluid adds to launch cost.

References I

- [1] Murat Bulut, Satish G Kandlikar, and Nedim Sozbir.
A review of vapor chambers.
Heat Transfer Engineering, 40(19):1551–1573, 2019.
- [2] Yogendra Joshi and Pramod Kumar, editors.
Energy Efficient Thermal Management of Data Centers.
Springer US, 2012.
- [3] Severin Zimmermann, Ingmar Meijer, Manish K Tiwari, Stephan Paredes, Bruno Michel, and Dimos Poulikakos.
Aquasar: A hot water cooled data center with direct energy reuse.
Energy, 43(1):237–245, 2012.